

Biomechanical Assessment of Human Movement After Total Ankle Replacement: A Case Study

Ari Friedman, Sierra Warnygora, Lauren Hanchar, and Daniel Ferris

University of Florida Human Neuromechanics Laboratory

Introduction

- Longitudinal case study of a male patient with end-stage osteoarthritis undergoing Total Ankle Arthroplasty (TAA)
- **Stryker Inbone II (4th Generation):** Designed to enhance stability and restore mobility during both normal and high-impact activities⁴
- Limited data on the long-term outcomes of 4th generation TAA systems²
- **Primary goal:** monitor the participant's recovery by analyzing gait *kinematics* and *kinetics*, with a focus on bilateral symmetry over time

PRE



POST



Methods

Participant

- 54-year-old male
- Recreational beach volleyball player

Testing Tasks

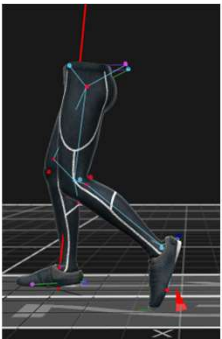


Data Collection Timeline



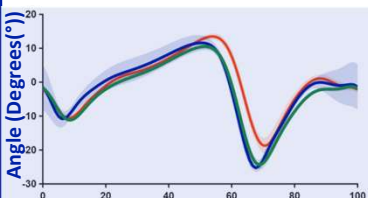
Data Collection

- Marker Based Motion Capture
- Electromyography (EMG)
- Bertec Force Plate Treadmill

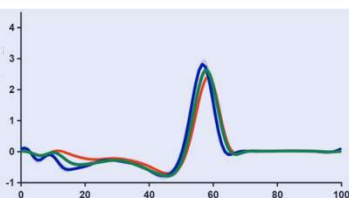
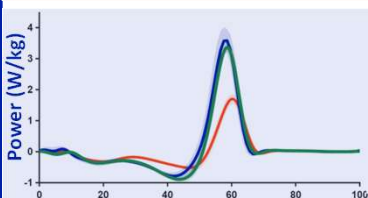
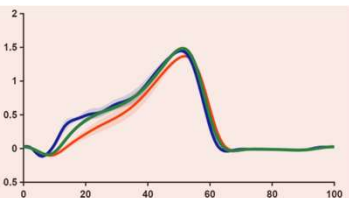
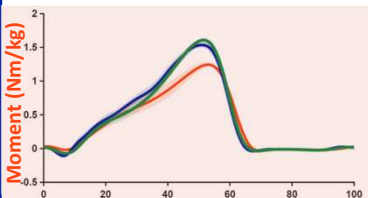
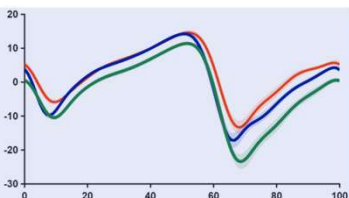


Results

Left Foot (Healthy)



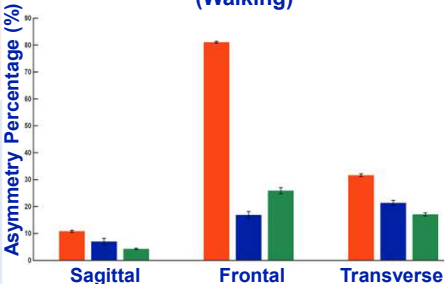
Right Foot (Injured)



Gait Cycle Percentage (%)

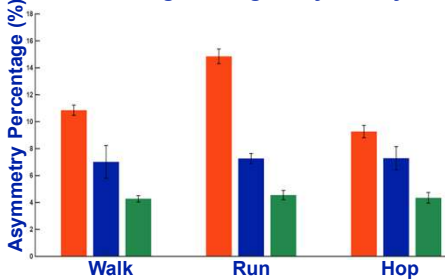
- Kinematic and kinetic measures show significant recovery by 12 months
- Bilateral symmetry improves, particularly in the sagittal plane
- Residual asymmetries in plantarflexor moment and power suggest incomplete restoration of propulsion

Ankle Angle Asymmetry (Walking)

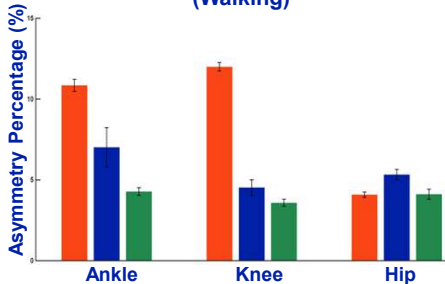


Asymmetry (%) — lower values indicate greater symmetry

Ankle Sagittal Angle Asymmetry



Joint Sagittal Angle Asymmetry (Walking)



Discussion

Symmetry Over Time:

- Bilateral symmetry improved from pre-surgery to 12 months, with the largest gains occurring early post-surgery and continued improvement over time

Task Differences:

- Walking, running, and hopping showed similar improvements in symmetry, indicating consistent recovery across task demands

Joint Behavior:

- The ankle demonstrated strong recovery, while the knee and hip suggested compensatory adaptations during the recovery process

Clinical Impact:

- Symmetry metrics may guide return-to-sport decisions and rehabilitation progression

References

- [1] Mercurio M, et al. Healthcare (Basel). 13(7):841. 2025
- [2] Easley ME, et al. J Am Acad Orthop Surg. 10(3):157-67. 2002
- [3] Queen RM, Dickerson LC, Ranganathan S, Schmitt D. J Biomech. 99:109531. 2020

- [4] Stryker. Inbone Total Ankle System. Stryker Foot & Ankle.
- [5] Kugach K, et al. Gait Posture. 113:272-9. 2024

Acknowledgements

Human Neuromechanics Laboratory, Dr. Miqi Wang, Dr. Daniel Ferris